NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

PERFORMANCE SPECIFICATION SHEET

FILTERS, RADIO INTERFERENCE,
STYLE FL64

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the filters described herein shall consist of this document and the latest issue of Specification MIL-F-15733.

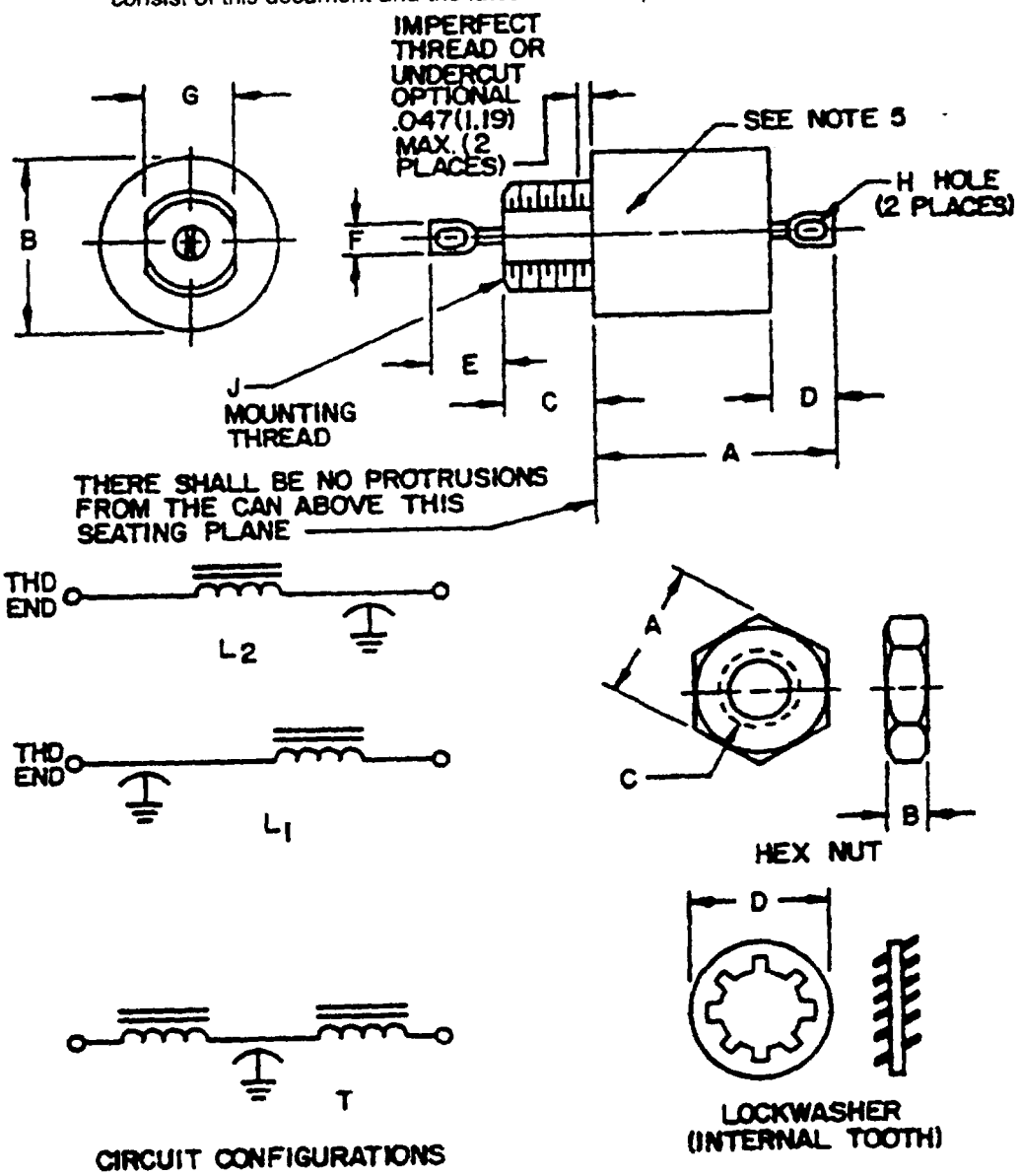


FIGURE 1. Filter and hardware dimensions, and circuit configurations

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TABLE I. Filter dimensions.

Dash No. 1/ 2/	A		B		C		D		E		F	G		H		J
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	TYP	Min	Max	Min	Max	
0001	---	.743 (18.87)	---	.416 (10.57)	.175 (4.45)	.205 (5.21)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)	.119 (3.02)	.190 (4.83)	.210 (5.33)	.045 (1.14)	.075 (1.91)	6-32 UNC-2A
0002	---	↓	---	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
0003	---	↓	---	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
0004	---	↓	---	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
0005	---	↓	---	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
0006	---	↓	---	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
0007	---	↓	---	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
0008	---	.743 (18.87)	---	.416 (10.57)	.175 (4.45)	.205 (5.21)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)	.119 (3.02)	.190 (4.83)	.210 (5.33)	.045 (1.14)	.075 (1.91)	6-32 UNC-2A

NOTES:
1/ For use in applications where mounting panel thickness does not exceed 0.062 inch.
2/ Recommended mounting torque: 48 ounce-inch, maximum.

FIGURE 1. Filter and hardware dimensions, and circuit configurations - Continued.

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TABLE II. Hardware dimensions.

Dash No.	Hex nut					Lockwasher	
	A		B		C	D	
	Min	Max	Min	Max	Thread	Min	Max
0001	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.410 (10.41)
0002	↓	↓	↓	↓	↓	↓	↓
0003	↓	↓	↓	↓	↓	↓	↓
0004	↓	↓	↓	↓	↓	↓	↓
0005	↓	↓	↓	↓	↓	↓	↓
0006	↓	↓	↓	↓	↓	↓	↓
0007	↓	↓	↓	↓	↓	↓	↓
0008	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.410 (10.41)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. All filters shall be supplied with mounting hardware.
5. Terminal identification (nonsymmetrical filters): The case shall be marked at the threaded end of the filter, with the symbol "C" or the symbol "L", as follows:

Circuit	Symbol
L_1	C
L_2	L

FIGURE 1. Filter and hardware dimensions, and circuit configurations - Continued.

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Type 1/ designation	Dash No.	Circuit configuration	Maximum rated current (amperes)	Maximum rated voltage DC +85°C to -55°C	Maximum rated voltage DC +125°C	Maximum rated voltage AC(Vrms) 60 Hz or 400 Hz +85°C to -55°C	Maximum rated voltage AC(Vrms) 60 Hz or 400 Hz +125°C	Voltage drop volts DC or volts rms max	Minimum full load insertion loss (dB) in accordance with MIL-STD-202 over full temperature range -55°C to +125°C: 30 kHz	50 kHz	100 kHz	150 kHz	200 kHz	300 kHz	500 kHz	1.0 MHz	2.0 MHz	5.0 MHz	10.0 MHz	20.0 MHz	50.0 MHz	100 MHz	1.0 GHz	10 GHz
FL64DW1ED3	0001*2/	L_2	1.0	200	200	150	150	.250	---	---	---	12	---	23	---	44	---	---	70	---	---	---	70	---
FL64DW1ED3	0002*3/	L_1	1.0	↑	↑	↑	↑	.250	---	---	---	12	---	23	---	44	---	---	70	---	---	---	70	---
FL64EW1ED3	0003*2/	L_2	3.0	↑	↑	↑	↑	.150	---	---	---	7	---	13	---	29	---	---	67	---	---	---	70	---
FL64EW1ED3	0004*2/	L_1	3.0	↑	↑	↑	↑	.150	---	---	---	7	---	13	---	29	---	---	67	---	---	---	70	---
FL64FW1ED3	0005*2/	L_2	5.0	↑	↑	↑	↑	.075	---	---	---	7	---	12	---	24	---	---	57	---	---	---	70	---
FL64FW1ED3	0006*2/	L_1	5.0	↑	↑	↑	↑	.075	---	---	---	7	---	12	---	24	---	---	57	---	---	---	70	---
FL64YW1ED3	0007*2/	T	2.0	↑	↑	↑	↑	.200	---	---	---	11	---	19	---	42	---	---	70	---	---	---	70	---
FL64WW1ED3 3/	0008*2/	T	4.0	200	200	150	150	.250	---	---	---	10	---	17	---	32	---	---	70	---	---	---	70	---

1/ The type designation does not describe a discrete item. For complete identification, the part number and type designation should be referenced.
2/ For applications where system voltage is 115 Vrms, 60 or 400 Hz.
3/ The current rating symbol "W" indicates the current rating specified in table III of this specification.
* Preferred for use in the design of equipment.

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REQUIREMENTS:

Dimensions and configuration: As specified in tables I and II and figures.

Case: Metal.

Seal: Glass to metal.

Terminals: Solderable (see figure 1).

Operating temperature range: -55°C to +125°C (temperature characteristic D).

Rated voltage: See table III.

Rated current: See table III.

Voltage drop: See table III.

Insertion loss characteristic: Symbol W (see table III).

Seal: Method 112, test condition A, MIL-STD-202.

Capacitance to ground: Method 305, MIL-STD-202.

Temperature rise: +7°C, maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202.
Test temperature: +25°C.
Test voltage: 2 times rated voltage (Vdc).
Duration of application of test voltage:
For qualification: 1.0 minute.
For quality conformance: 1.0 to 5.0 seconds.
Points of application of test voltage: Between the case (ground) and all terminals connected together.
Maximum charging and discharging current: 50 mA.

Barometric pressure (reduced): Method 105, test condition D, MIL-STD-202.
Dielectric withstanding voltage: In accordance with initial requirements except that the test voltage shall be 1.25 times the rated voltage.

Inrush transients test: The filters shall be subjected to test by instantaneous application of 115 Vrms, 400 Hz to the filter with the output unloaded. The source impedance shall be less than 0.5 ohm and its energy capability shall not limit the inrush current to the filter. The filters shall be subjected to this inrush a minimum of 10 times, with a maximum time of 5 minutes between applications.
Measurements after test: Insulation resistance (at +125°C) and insertion-loss (check test) shall meet initial requirements.
The input shall always be into the inductor for non-symetrical filters.

Transient voltage tests:
MIL-STD-461 transients: The filters shall be subjected to transients as defined by category CS06 of MIL-STD-461, figure 19, for a category B, 115 Vrms, 400 Hz system. The test method and test equipment is as defined in MIL-STD-462 for the same category. The filter output load shall be an open circuit.
Measurements after test: Insulation resistance (at +125 C) and insertion-loss (check test) shall meet initial requirements.

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MIL-STD-704 transients: The filters shall be subjected to transient tests to represent figure 3 (category B system) of MIL-STD-704. This shall be accomplished as follows:
Preconditioning: Filter shall be energized at 115 Vrms, 400 Hz carrying rated current for 15 minutes minimum.
Transient application: The output load shall be removed and the input voltage shall be stepped form 115 Vrms to 180 volts within 20 milliseconds of removal of the load. The step from 115 to 180 volts shall be accomplished with a rise time of less than 60 milliseconds. The 180 volts shall be maintained for 100 milliseconds minimum. This cycle shall be repeated 10 times.
Measurements after test: Insulation resistance (at +125°C) and insertion-loss (check-test) shall meet initial requirements. At the option of the manufacturer, the measurements after MIL-STD-461 transient tests may be postponed until after MIL-STD-704 tests.

Insulation resistance: Method 302, MIL-STD-202.
Test potential: Rated dc voltage.
Maximum charging and discharging current: 50 mA.
Insulation resistance: Not less than 100 megohms at +25°C and not less than 5 megohms at +125°C, when measured between the case (ground) and all terminals connected together.

Voltage drop: In accordance with MIL-F-15733. Maximum voltage drop at +25°C shall be as specified in table III.

Insertion loss: In accordance with MIL-F-15733 and table III.

Overload: In accordance with MIL-F-15733.
Measurements at +25°C after test: Insulation resistance, insertion-loss (check test) and voltage drop shall meet initial requirements.

Terminal strength: Method 211, test condition A (pull), MIL-STD-202.
Applied force: 4.5 pounds.
Measurements after test: Insulation resistance, insertion-loss (check test), and voltage drop shall meet initial requirements.

Flash point of impregnant or potting compound: Not applicable.

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.
Depth of immersion: To within .063 (±.032) of the insulation.
Measurements at +25°C after test: Insulation resistance, insertion-loss (check test), and voltage drop shall meet initial requirements.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Thermal shock and immersion:
Thermal shock: Method 107, test condition A, MIL-STD-202, except the temperature at step 3 shall be +125 °C.
Immersion: Method 104, test condition A, MIL-STD-202.
Measurements after final cycle: In accordance with MIL-F-15733.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.
Electrical load: Rated ac voltage.
Rated current: See table III.
Point of application: Between case (ground) and one input terminal.
Measurements after test: Insulation resistance, insertion-loss (check test), and voltage drop shall meet initial requirements.

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Vibration, high frequency: Method 204, test condition D (20 G), MIL-STD-202.
Electrical load:
Voltage: Rated ac voltage.
Current: As specified in table III.
Measurements after test: Insulation resistance, insertion-loss (check test) and voltage drop shall meet initial requirements.

Moisture resistance: Method 107, test condition A, MIL-STD-202, except that the temperature at step 5 shall be +125°C. No measurements shall be made before or after cycling.
Filters shall then be tested in accordance with method 106, MIL-STD-202.
Initial measurements: Not applicable.
Polarization voltage: Rated dc voltage.
Measurements after 24-hour drying period:
Dielectric withstanding voltage: Voltage at 1.8 times rated dc voltage.
Insulation resistance and insertion-loss (check test) shall meet initial requirements.

Solderability: Method 208, MIL-STD-202.
Number of terminations to be tested: 2
Special preparation of terminations: None
Application of standard solderable wire: Start through terminal hole with a minimum of 1.5 full turns wrap around terminal.
Depth of immersion: To within .063 (±.032) of insulation.

Life: Method 108, MIL-STD-202. Test condition D (1,000 hours) for qualification inspection, test condition B (250 hours) for group C inspection.
Test temperature: +125°C.
Test voltage: 1.2 times rated ac voltage, 400 Hz.
Test current: Maximum rated current (see table III)
Measurements after test at +25°C:
Dielectric withstanding voltage at 1.8 times rated dc voltage.
Insulation resistance and insertion-loss (check test) shall meet initial requirements.

Voltage conditioning: Voltage conditioning shall consist of applying rated ac voltage 400 Hz to the part at +125°C. The voltage shall be applied for a minimum of 250 hours. Charging and discharging current shall be limited within the range of 30 to 150 milliamperes. The test circuit shall be in accordance with figure 2. E_1 shall be continuously monitored and if it falls below 95% of E_T, the test shall be stopped and the defective part removed. Time counted toward total test time shall be the time that E_1 is greater than 95% of E_T. After the completion of 250 hours and while remaining at +125°C, the insulation resistance shall be measured and shall be 5 megohms minimum.

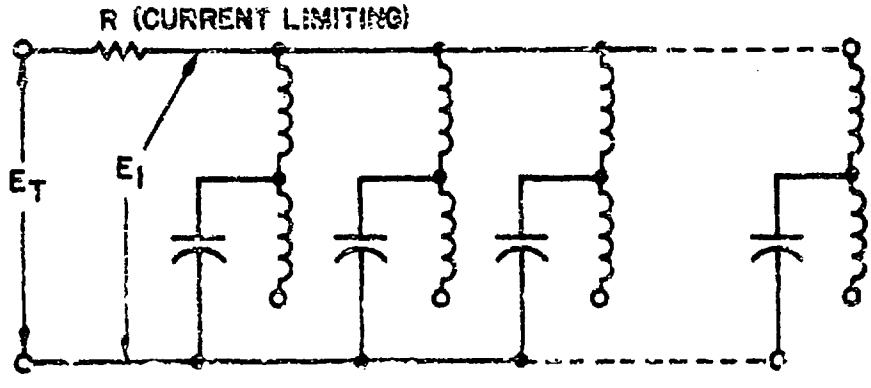


FIGURE 2. Voltage conditioning test circuit.

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QUALITY ASSURANCE PROVISIONS

Quality assurance provisions shall be as specified in MIL-F-15733 except group A inspection shall be as specified in table V of this specification, in the order shown. Lots which have greater than 5% rejects shall be considered reject lots. Reject lots may, at the option of the supplier, be resubmitted to 100% group A inspection; however, resubmitted lots which have greater than 2% rejects shall be rejected any may not be resubmitted for reinspection.

Requirements of this specification not presently required in MIL-F-15733 shall be included in qualification inspection as group V and in group C tests. These are shown in tables IV, VI, and VII of this specification.

TABLE IV. Qualification inspection. 1/

Examination or test	Number of samples to be inspected	Number of defectives allowed
Group V		
Transient voltage	4	0
Inrush transients	4	
Solderability	4 (any of above 8)	1

1/ This table is an addition to table IX of qualification inspection.

TABLE V. Group A inspection.1/

Examination or test	AQL (percent defective) or 100% inspection	
Thermal shock	100%	
Voltage conditioning	100%	
Dielectric withstanding voltage	100%	
Insulation resistance	100%	
Voltage drop	100%	
Seal test	100%	
Insertion-loss (check test)	100%	
	Major	Minor
Visual and mechanical of MIL-F-15733	1.0	4.0
Capacitance to ground	1.0	4.0

1/ This table replaces table X of MIL-F-15733.

TABLE VI. Group B inspection. 1/

Examination or test	Requirement paragraph	Method paragraph
Barometric pressure (reduced)	3.10	4.6.6

1/ This table is in addition to table XI of MIL-F-15733.

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TABLE VII. Group C inspection. 1/

Examination or test	Number of samples to be inspected	Number of defectives allowed
Transient voltages Inrush transients Solderability	4 4 4 (any of above 8)	0 1

1/ This table is in addition to table XII of MIL-F-15733.

Supersession data: This specification includes requirements of contractor 13160 Drawing No. 7890396.

Part number: M15733/59-(dash number from table I).

Preparing activity:
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Agent:
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(Project 5915-F109-2)

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